

PREDICT. EXPLAIN. PREVENT.

YieldSentinel AI

AI-Powered Wafer Intelligence Platform

ThinkTankX

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THE PROBLEM

A Failed Wafer Is Easy to Detect. Finding WHY Is Hard.

Thousands of Sensors



High-Volume, Noisy Telemetry



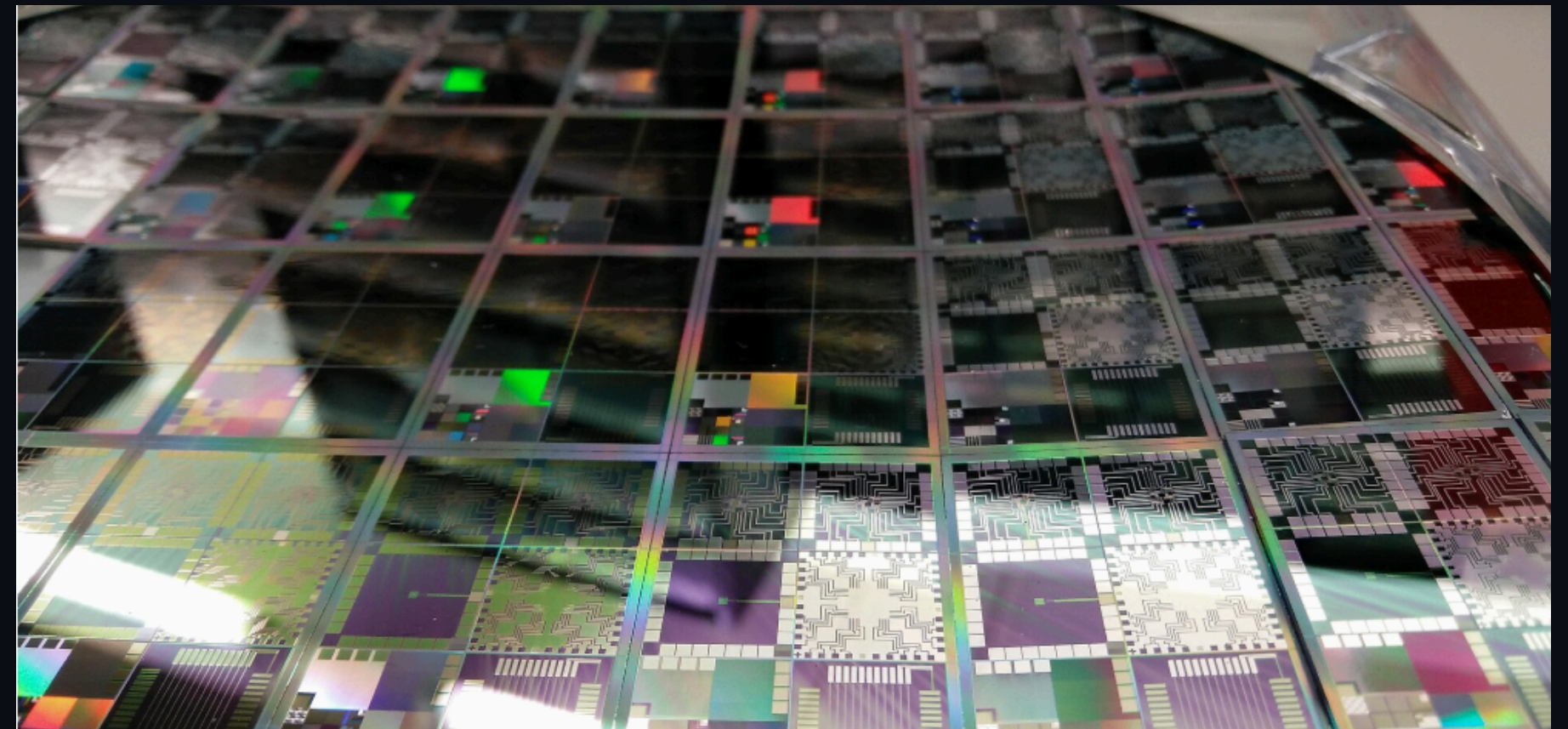
Many Possible Causes



Slow Manual Investigation



Yield & Revenue Loss



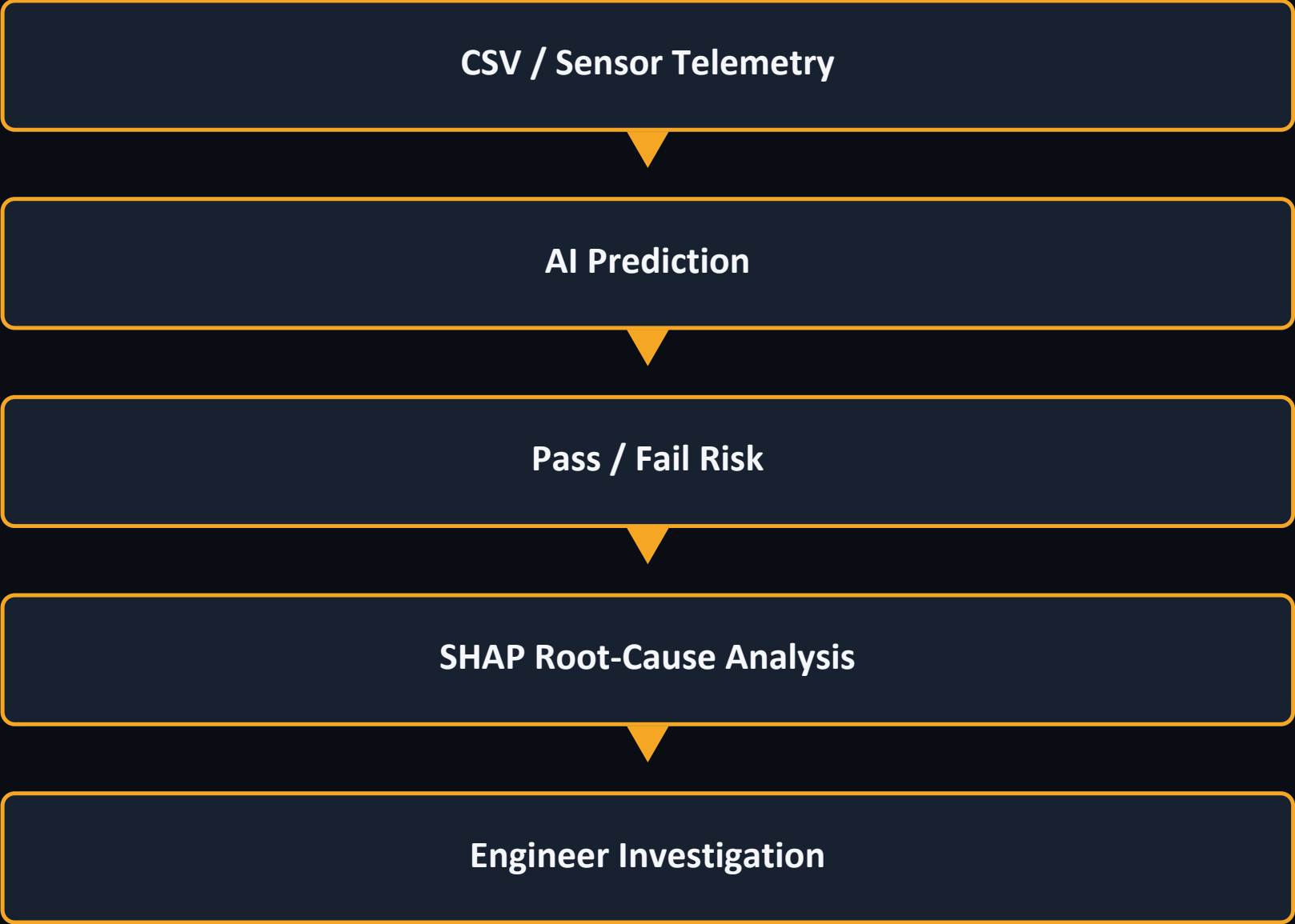
- High-dimensional sensor data
- Too many possible failure causes
- Slow, inconsistent root-cause analysis

WHO FACES THIS: Quality Engineers • Process Analysts • Production Leaders

THE SOLUTION

Meet YieldSentinel AI

Turn complex wafer telemetry into prediction + explanation + action.



PREDICT

Pass / Fail probability per wafer

EXPLAIN

Rank sensors contributing to failure

VISUALIZE

Yield, risk & defect patterns



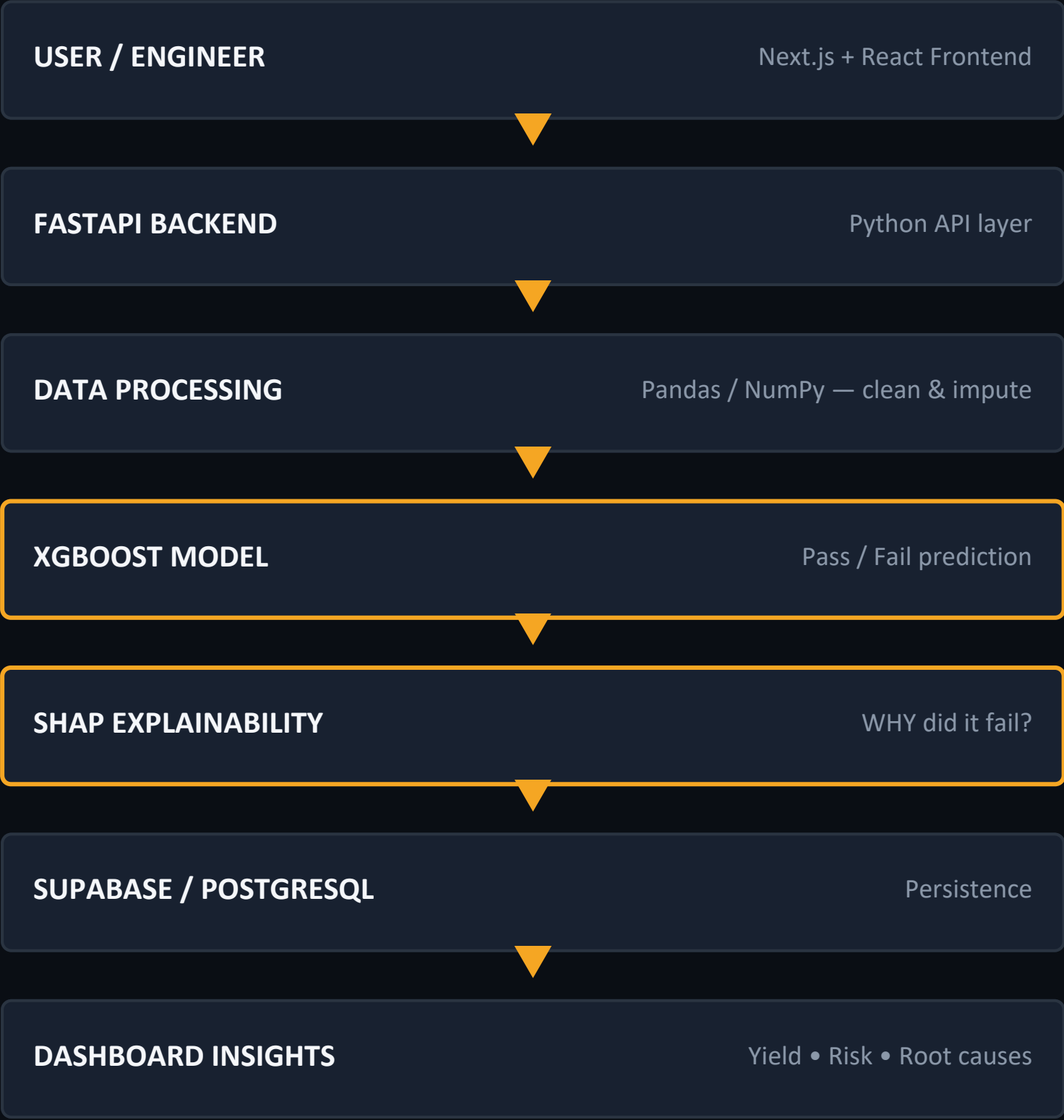
COMMAND CENTER — LIVE DASHBOARD

Implemented System

- XGBoost prediction
- SHAP-based explanation
- Defect-pattern visualization
- AI assistant

HOW IT WORKS

From Wafer Data → Engineering Insight



AI ASSISTANT BRANCH



Not IBM Bob — the in-app chatbot runs on OpenRouter / NVIDIA Nemotron.

TECHNOLOGY STACK

FRONTEND

Next.js, React, TailwindCSS

ML

XGBoost

DATABASE

Supabase / PostgreSQL

BACKEND

FastAPI, Python, Pandas, NumPy

EXPLAINABILITY

SHAP

LLM

NVIDIA Nemotron via OpenRouter

DEMO / KEY FEATURE

The "WHY?" Engine



Batch Prediction (CSV)

Process multi-parameter wafer sensor logs through high-dimensional anomaly detection models.

Model Engine Ready: XGBoost-SECOM-v2.8 & Root Classifier Active

WEIGHTS: SECOM_1549.OTS

Upload Wafer Data

Required: 502 features

Upload a CSV containing the required model features to analyze this batch for defect trends and parameter drifting.

Drop CSV here or click to browse

Requires the 502 model sensor columns (SECOM format)

Ready for tensor execution

WAFERS: 1

SENSORS: 500

FORMAT: UTF-8 CSV

Batch Summary

STAGED

Preliminary telemetry verification for the currently staged dataset prior to model inference.

Input Schema Match

100% Validated

590 / 590 Passed

0 Missing Columns

Estimated Execution Time

~ 2.4 Seconds

GPU Tensor Acceleration

Batch Size: 256

TARGET FAB CLUSTER

CLEANROOM-ALPHA // 30M

Cluster Capacity: 80%

Load: Nominal

INSPECTION PIPELINE

Sensor Normalization (Z-Score)

Spatial Defect Clustering

Batch Prediction (CSV Upload)

DATA ANALYST

Your uploaded analyses — click Load to make one active for all analysis pages

REFRESH

DATASET	DATE	WAFERS	YIELD	STATUS	ACTIONS
uci-secom.csv	Sep 15, 26	1567	91.5%	ACTIVE	LOADED DELETE
uci-secom_synthetic.csv	Sep 15, 26	1567	87.1%	INACTIVE	LOAD DELETE
uci-secom.csv	Sep 15, 26	1567	91.5%	INACTIVE	LOAD DELETE

Wafer Predictions

EXPORT CSV

#	WAFER ID	STATUS	FAIL PROBABILITY	PASS PROBABILITY
1	WAFER-1	PASS	9.87%	90.13%
2	WAFER-2	PASS	6.75%	93.25%
3	WAFER-3	FAIL	95.15%	4.85%
4	WAFER-4	PASS	3.98%	96.02%
5	WAFER-5	PASS	5.93%	94.07%
6	WAFER-6	PASS	18.93%	81.07%
7	WAFER-7	PASS	18.38%	81.62%
8	WAFER-8	PASS	18.53%	81.47%

Wafer Predictions

POWER BY-STYLE COMMAND CENTER

Yield Intelligence

Cross-analysis intelligence — trends, risks, root cause evolution, and process status.

YIELD TRENDS

91.51%

YIELD TRENDS

87.11%

YIELD TRENDS

4.48%

YIELD TRENDS

3

YIELD TRENDS

4701

YIELD TRENDS

5

PROJECT STATUS

IMPROVING

YIELD TRENDS

YIELD TRENDS

YIELD TRENDS

CURRENT TOP RISKS

Sensor 103

Sensor 10

Sensor 100

Sensor 300

Sensor 400

CRACK STATUS

Blocked

Improving

Partially

New Risk

Regression

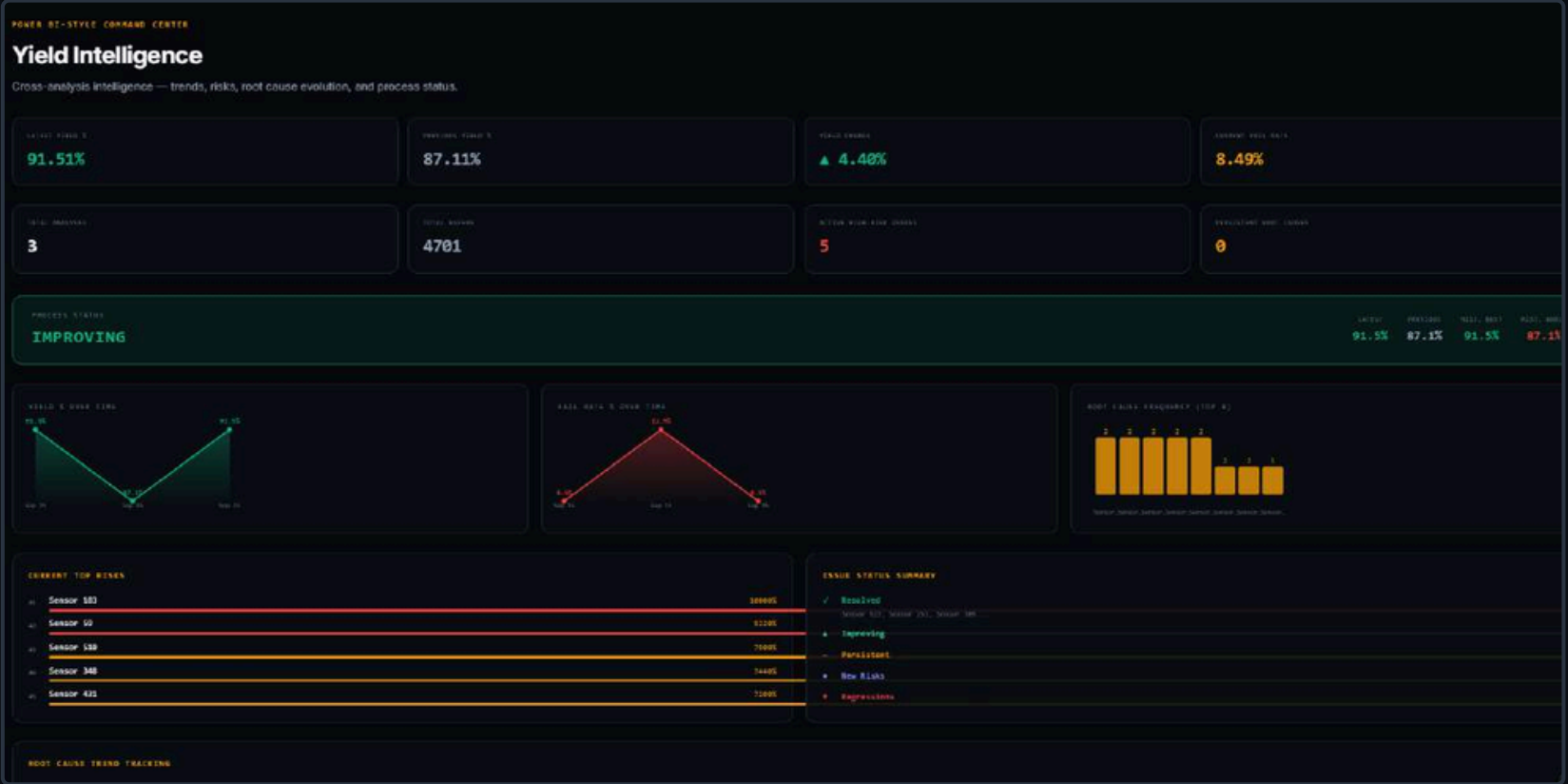
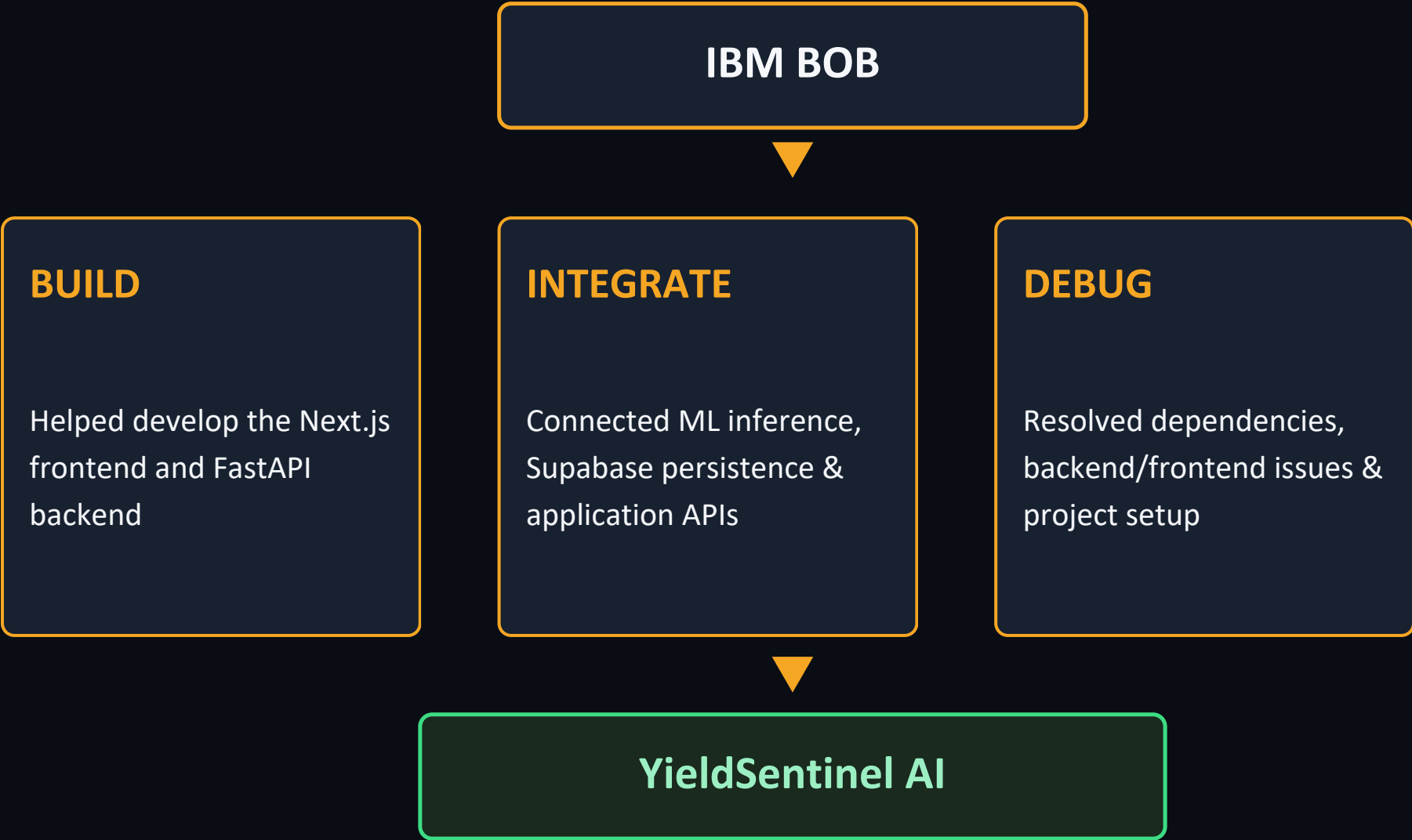
ROOT CAUSE TREND TRACKING

Root Cause Ranking

DEMO FLOW: Upload CSV → Yield → Root Causes → Defect Patterns → AI Assistant

IBM TECHNOLOGIES

Built with IBM Bob



WHAT BOB IS NOT

- ✗ Not the user-facing chatbot (that's OpenRouter / NVIDIA Nemotron)
- ✗ Not a source of wafer predictions
- ✗ Not a production monitoring system

"IBM Bob acted as our AI development partner from idea to a working full-stack product."

From Raw Telemetry to Decisions



Fast Batch Analysis

Process large wafer datasets quickly

Explainable Predictions

Identify the sensors contributing to risk

Decision-Ready Dashboard

Surface yield, risk & root-cause information

DEMONSTRATED

- ✓ Rapid CSV ingestion + prediction
- ✓ SHAP-based sensor ranking
- ✓ Batch-level yield analytics

Based on actual implemented capabilities — no invented metrics.

EXPECTED IMPACT (not yet measured)

Earlier visibility → Faster investigation → Reduced downstream yield loss

No accuracy, F1, precision, recall, or cost-savings figures are claimed — none have been formally measured.

WHAT'S NEXT

From Detection to Prevention

TODAY

CSV Telemetry

Prediction

Root-Cause Explanation

Engineer Action

FUTURE

Real-Time Fab Data Streams

Continuous Risk Detection

Root-Cause Explanation

Preventive Action

Real-Time Fab Streams

Equipment / fab data feeds

Spatial Defect Analysis

True die-map defect visualization

Enterprise Auth / SSO

Secure multi-user access

THINKTANKX

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"Don't just detect failure. Explain it."